Midterm 2 Free Response Solutions (By Problem)

Contents

Free Response Solutions (By Problem)

Question 22

Part (a): Absolute Advantage (Burgers)

Version A **Linda** has the absolute advantage because she can produce **40 burgers** per shift, which is more than Bob's **20 burgers**.

Version B **Linda** has the absolute advantage because she can produce **30 burgers** per shift, which is more than Bob's **10 burgers**.

Version C **Linda** has the absolute advantage because she can produce **50 burgers** per shift, which is more than Bob's **25 burgers**.

Version D **Linda** has the absolute advantage because she can produce **20 burgers** per shift, which is more than Bob's **15 burgers**.

Version E **Linda** has the absolute advantage because she can produce **60 burgers** per shift, which is more than Bob's **30 burgers**.

Part (b): Bob's Opportunity Cost of 1 Burger

Version A Bob gives up 60 cookies to make 20 burgers. Calculation: $\frac{60 \text{ cookies}}{20 \text{ burgers}} = 3$ cookies per burger

Version B Bob gives up 40 cookies to make 10 burgers. Calculation: $\frac{40 \text{ cookies}}{10 \text{ burgers}} = 4$ cookies per burger

Version C Bob gives up 50 cookies to make 25 burgers. Calculation: $\frac{50\text{ cookies}}{25\text{ burgers}}$ = 2 cookies per burger

Version D Bob gives up 60 cookies to make 15 burgers. Calculation: $\frac{60\text{ cookies}}{15\text{ burgers}}$ = 4 cookies per burger

Version E Bob gives up 90 cookies to make 30 burgers. Calculation: $\frac{90\text{ cookies}}{30\text{ burgers}}$ = 3 cookies per burger

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Part (c): Comparative Advantage (Cookies)

Version A **Bob** has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{3}$ burger (20/60), while Linda's OC is $\frac{1}{2}$ burger (40/80).

Version B **Bob** has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{4}$ burger (10/40), while Linda's OC is $\frac{1}{3}$ burger (30/90).

Version C **Linda** has the comparative advantage. Linda's OC of 1 Cookie is $\frac{1}{3}$ burger (50/150), while Bob's OC is $\frac{1}{2}$ burger (25/50).

Version D **Bob** has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{4}$ burger (15/60), while Linda's OC is $\frac{1}{2}$ burger (20/40).

Version E **Bob** has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{3}$ burger (30/90), while Linda's OC is $\frac{1}{2}$ burger (60/120).

Part (d): Range of Prices (Terms of Trade)

Version A Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 3 cookies). Range: **Between 2 and 3 cookies per burger**.

Version B Trade will occur if the price of 1 burger is between the two workers' opportunity costs (3 and 4 cookies). Range: **Between 3 and 4 cookies per burger**.

Version C Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 3 cookies). Range: **Between 2 and 3 cookies per burger**.

Version D Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 4 cookies). Range: **Between 2 and 4 cookies per burger**.

Version E Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 3 cookies). Range: **Between 2 and 3 cookies per burger**.

Question 23

Part (a): Initial Total Cost

Version A Combined production cost for the first 6 widgets: \$10 + \$25 + \$15 + \$30 + \$20 + \$50 = **\$150**.

Version B Combined production cost for the first 6 widgets: \$12 + \$20 + \$16 + \$32 + \$22 + \$48 = **\$150**.

Version C Combined production cost for the first 6 widgets: \$15 + \$30 + \$20 + \$35 + \$25 + \$60 = **\$185**.

Version D Combined production cost for the first 6 widgets: \$20 + \$30 + \$25 + \$35 + \$30 + \$65 = **\$205**.

Version E Combined production cost for the first 6 widgets: \$5 + \$15 + \$10 + \$20 + \$15 + \$40 = **\$105**.

Part (b): WTP and WTA

Version A

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$60 - \$40 = **\$20**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$60 - \$50 = **\$10**.

Version B

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$60 - \$38 = **\$22**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$60 - \$48 = **\$12**.

Version C

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$70 - \$45 = **\$25**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$70 - \$60 = **\$10**.

Version D

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$80 - \$40 = **\$40**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$80 - \$65 = **\$15**.

Version E

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$50 - \$25 = **\$25**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$50 - \$40 = **\$10**.

Part (c): After-Trade Total Cost

Version A Cost is minimized by using the 6 permits on the lowest marginal costs available: 10, 15, 20, 25, 30, and 40. New Total Cost: **\$140**.

Version B Cost is minimized by using the 6 permits on the lowest marginal costs available: 12, 16, 20, 22, 32, and 38. New Total Cost: **\$140**.

Version C Cost is minimized by using the 6 permits on the lowest marginal costs available: 15, 20, 25, 30, 35, and 45. New Total Cost: **\$170**.

Version D Cost is minimized by using the 6 permits on the lowest marginal costs available: 20, 25, 30, 30, 35, and 40. New Total Cost: **\$180**.

Version E Cost is minimized by using the 6 permits on the lowest marginal costs available: 5, 10, 15, 15, 20, and 25. New Total Cost: **$90**.